# Improving fatigue performance of porous orthopaedic implants through manufacturing optimisation

**D. Giannetto[1], S. Ghouse[2], A. López-Barberena[1], S. Martin[3], A. Veiga[3], , N. Rodriguez-Florez[1,4], S. Ruiz de Galarreta[1]**

*[1] TECNUN-School of Engineering, University of Navarra, Spain, dgiannetto@unav.es*
*[2] Imperial College London, United Kingdom*
*[3] CEIT-Basque Research and Technology Alliance (BRTA), Spain*
*[4] Ikerbasque, Basque Foundation for Science, Spain*

## Introduction

Load-bearing orthopaedic implants require porous architectures that promote bone ingrowth while maintaining mechanical integrity under cyclic loading. Ti-6Al-4V gyroids manufactured by laser powder bed fusion (PBF-LB/M) are promising candidates, owing to their interconnected porosity and favourable mechanical response. However, their fatigue performance under physiological loading is limited by surface irregularities, geometrical deviations, manufacturing defects and the martensitic microstructure formed by rapid solidification [1,2]. This study evaluates two complementary manufacturing strategies to improve the fatigue performance of Ti-6Al-4V gyroids: geometrical accuracy through local process-parameter optimisation and heat treatments to improve the microstructure.

## Methods

To optimise geometrical accuracy, inclined thin-walls were produced by PBF-LB/M with different laser power and scan-speed combinations and assessed by µCT to identify optimal process windows. The resulting process windows were implemented in an inclination-aware workflow assigning laser parameters according to surface inclination (Figure 1). Optimised gyroids were compared with structures using manufacturer's standard parameters.

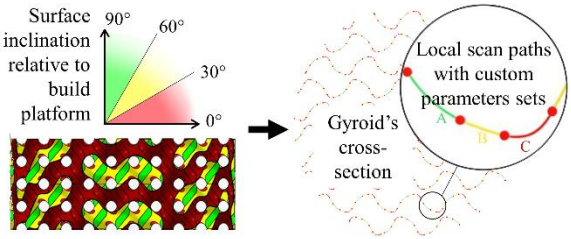



*Figure 1 – Gyroid laser scan paths tailored to local surface inclination.*

To investigate the effect of microstructural optimisation on fatigue performance, cylindrical gyroid specimens were manufactured using standard process parameters and divided into four conditions: as-built and vacuum heat-treated at 600, 850, or 1050 °C. Microstructure was examined by SEM, and compression-fatigue tests were conducted under load control at 15 Hz.

## Results

Scan speed and laser power governed geometrical quality and mechanical performance. The inclination-aware workflow improved thickness uniformity, reduced crack-initiating surface irregularities, and increased compressive strength compared with the manufacturer's standard parameters. Heat treatment progressively transformed the fine acicular α' martensitic microstructure into α+β lamellar structures. All heat-treated groups outperformed the as-built specimens in fatigue, with the 1050 °C condition showing the best high-cycle behaviour.

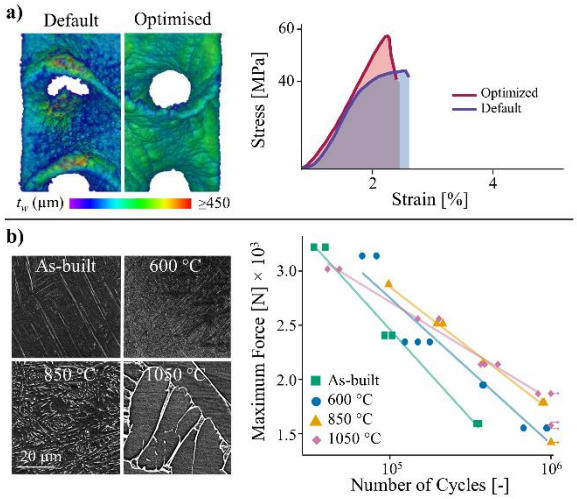



*Figure 2 – (a) Thickness ($t_w$) and compressive response following process-parameter optimisation; (b) effect of microstructural optimisation on fatigue performance.*

## Conclusions

Fatigue performance of porous Ti-6Al-4V gyroids is governed by both geometrical quality and microstructural state. Locally adapting laser parameters can reduce defect-prone surface features, while heat treatments improve fatigue resistance through martensite decomposition and residual-stress relaxation. Combining both strategies offers a promising route towards long-term mechanical performance of orthopaedic implants.

## References

[1] Vrancken et al, J Alloys Compd, 541:177-185, 2012.
[2] Kelly et al, Acta Biomater, 94:610-626, 2019